

Clinical Applications and Existing Problems of Medical Robots

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Abstract - Medical robots that integrate artificial intelligence and robotics technology are changing the traditional medical models. The International Federation of Robots divided them into four categories of robots such as: surgical robots, rehabilitation robots, assistive robots, service robots. Surgical robots are widely used in various departments of medicine; rehabilitation robots serve the patients' recovery in various departments; Assistive and service robots contribute to the efficiency and quality of medical services. Now medical robot still face the technological bottlenecks and clinical application difficulties; however, it has very broad application prospect and will serve medical services even more importantly in future.

Keywords - Medical robots; Surgical robots; Rehabilitation robots

I. INTRODUCTION

As a profound integration of artificial intelligence and robotics technology in the medical field, medical robots are gradually transforming traditional medical models and enhancing the efficiency and quality of medical services. The industrial scale of medical robots in the international market grew from \$11.9 billion in 2017 to \$20.7 billion in 2021, representing an increase of 73.9%. Meanwhile, the industrial scale of medical robots in the domestic market expanded from \$380 million in 2017 to \$1.13 billion in 2021, marking a growth of 197.4%.[1] The following will systematically review the current application status of medical robots from multiple dimensions.

II. APPLICATION FIELDS OF MEDICAL ROBOTS

Medical robots demonstrate broad application prospects across multiple fields. The International Federation of Robotics (IFR) categorizes medical robots into four major types: surgical robots, rehabilitation robots, assistive robots, and service robots [2].

A. Surgical Robots

Surgical robots are the most developed and wideused type of medical robots. They can filter out the natural shaking of a doctor's hand, reduce the probability of operation accidents, improve the accuracy and efficiency of operations. China's medical robot for surgery in industry scale had achieved 5.25 billion yuan in 2021, which took up 66.5% share of the total size of China's medical robot industry.[1] The da Vinci Surgical System by Intuitive Surgical [3] applies machine

learning algorithm to optimize the trajectory of motion of robot arms. Its function of filtering tremors is designed to enable suturing with a sub-millimeter accuracy, and already millions of surgical procedures have been performed by robots all over the world.

Surgical robots have been widely utilized in neurosurgery, orthopedic surgery, laparoscopic and minimally invasive surgery, vascular interventional surgeries or natural orifice surgery etc

1) Neurosurgery

Neurosurgical robots, based on medical imaging technology such as MRI, CT to construct 3D reconstruction, to allow doctors to intuitively make surgical path planning. Meanwhile, through the precise movements of robotic arms, they reduce human errors [4]. Example: The neurosurgical robot of Huake Precision adopts the frameless stereotactic brain with the accuracy up to 1mm. It can be used for the accuracy operations of brain tissue puncture, tumor biopsy, Deep Brain Stimulation (DBS) electrode implantation and so on based on the navigation. The navigation integrates high precision robot arm, 3-D positioning navigation and intelligent environment obstacle model that greatly ensure operation safety and success ratio.

2) Orthopedics

Orthopedic surgical robots can realize all-the-way automation from preoperative plan to intraoperative process through combining imaging technology, mechanical technology and robot technology, effectively lowering risks associated with surgery. For instance, "Orthopedic Navigation Robot" developed by Tianzhihang and pelvic fracture reduction robot developed by RosenBot are two different directions. Tianzhihang robot emphasizes "all-round coverage", improving the efficiency of general orthopedic surgeries, which realizes general operations, while RosenBot robot emphasizes "complicated fracture reduction", transcending the constraints of conventional surgery and meets the unresolved clinical demands. Two robots realize sub-millimeter operation precision by intelligent technologies, Tianzhihang realizes joint replacement and spinal operation, RosenBot is specialized in accurate treatment of pelvic fracture reduction, the two robots' operation realizes significant improvement on operation accuracy and patient recovery rate by pre-operation planning, intra-operation navigation, force feed-back technology [5].

3) Laparoscopic and Minimally Invasive Surgery

Laparoscopic and minimally invasive surgical robots perform complex surgeries through small incisions [6-7], reducing the risks of bleeding and infection and shortening hospital stays. For instance, laparoscopic surgical robots represented by the da Vinci Surgical System [8] are widely used in minimally invasive surgeries such as prostatectomy and hysterectomy. Their highly flexible robotic arms and three-dimensional high-definition vision enable more delicate surgical operations and less trauma to patients.

4) *Vascular Intervention*

Vascular intervention surgical robots combine ultrasound imaging and robotic technology to achieve radiation-free, high-precision vascular puncture. For example, an ultrasound-guided femoral vein puncture robot guides the puncture needle into the femoral vein in real-time through ultrasound imaging, improving the success rate and safety of punctures. This system is suitable for patient groups such as children, the elderly, and those with difficult-to-identify veins [9].

5) *Natural Orifice Surgery*

Natural orifice surgical robot combines the functions of flexible robotic arms with high definition imaging to operate in small diameter natural passageways, such as single-arm single-port laparoscopic surgical robot, performs surgery through the human's natural passageway (e.g., airway) and the abdominal cavity, to minimize more surgical trauma. For example, bronchoscope surgical robots are applied in the peripheral pulmonary nodules biopsies, and single-arm single-port laparoscopic surgical robots are applied in the cholecystectomy surgery etc.

B. *Rehabilitation Robots*

Rehabilitation robots are applied by bidirectional feedback of neural system to help patients recover and are divided into several types, such as functional replacement and assistive robots. It plays an important role in helping patients with recovery and improving the quality of their life. As one key part of medical robots, rehabilitation robot have been extensively used for the multiple departments.

1) *Neurological Rehabilitation Department*

The neurological rehabilitation robots use multi-degree-of-freedom robot arms to flexibly imitate the entire range of motion and combinations of upper limb motions, using sensor equipment to record and supervise a patient's status of motor situation in real-time, and returning the detailed report of the assessment of a rehabilitation system to a doctor. For example, the HUST-IR Upper Limb Rehabilitation Medical Robot is aimed at those patients with neurological injury and can allow them to recovery the function of upper limbs through emulation of daily upper limbs' motions. Such a robot combines force feedback and virtual reality technology with personalized rehabilitation training programs.

2) *Orthopedic Rehabilitation Department*

Orthopedic rehabilitation robots utilize three-dimensional gait analysis and mechanical feedback technologies to accurately assess patients' gait characteristics and provide targeted rehabilitation training while ensuring the safety of the

training process. For example, the user behavior-based lower limb rehabilitation robot aids post-orthopedic surgery patients in regaining walking ability through intelligent gait training [10]. This robot can automatically adjust training intensity based on patients' rehabilitation progress, improving rehabilitation efficiency.

3) *Geriatric Rehabilitation Department*

Geriatric rehabilitation robots should maintain the lightweight concept and human factors principles so as to guarantee the user comfort and safety in usage. At the same time, they can be made combined with the analysis of big data to offer a customized rehabilitation training scheme for patients. Take the upper limb rehabilitation medical robot based on user demand for example, it is aimed to elderly stroke or arthritis patients who are susceptible to upper limb dysfunction, it can motivate patients to improve their rehabilitation by gamified rehabilitation training interface. Also, the remote monitoring function of this robot enables doctors to adjust rehabilitation plans at any time.

4) *Pediatric Rehabilitation Department*

Pediatric rehabilitation robots are oriented towards entertaining, interactivity design, which can attract the children's attention. While taking children's development and growth characteristics into consideration, it also uses changable mechanical structure to adapt to different children's age's requirement. For instance, the children's cerebral palsy or developmental delay can be solved by rehabilitation robots that they train the children by interesting games to arouse their rehabilitation interest. Exoskeleton robots, for instance, can be used for assisting children during walking training to enhance the walking function.

C. *Assistive and Service Robots*

Assistive robot in medical field is widely used more and more extensive range from diagnosis and medical treatment to rehabilitation, improve the efficiency and quality of medical services.

1) *Surgical Departments*

Assistive surgical robot involves the high accuracy of motion control technology, real-time image navigation technology and the technology of force feedback which helps to complete a more delicate and more secure surgical operation. For instance, the interventional medical robot controlled by magnets, which can be widely used in cardiovascular surgery, neurosurgical and other fields. The assistive surgical medical robot can be widely used in the field of neurosurgery. Magnet-guided interventional medical robots control the trajectory of invivo micro-robots in external magnetic fields with precision so as to accomplish vascular interventional therapies in a non-invasive or minimally invasive manner. Assisted surgical medical robots (for example, da Vinci Surgical System) have been extensively used in various surgical fields with high degrees of surgical precision and safety.

2) *Rehabilitation Departments*

Assistive rehabilitation robots contain multi-degree-of-freedom robotic arm, force/position feedback system and tailor-made rehabilitation training program,

ensuring safe and comfortable rehabilitation. Take upper and lower limb rehabilitation robot for example, base on user's behavior and needs, it can restore their limbs function by mimicking natural motion patterns [10]. Such robots integrate sensors and personalized rehabilitation program to bring out the highly specialized training for rehabilitation.

3) *Nursing Departments*

Assistive nursing robots possess functions such as autonomous navigation, intelligent obstacle avoidance, voice interaction, and remote monitoring, improving the efficiency and quality of nursing services. For example, medication delivery and patient ward-round robots, as well as autonomous following medical assistive robots, have been widely used in nursing. Medication delivery and patient ward-round robots can automatically complete drug delivery [11] and patient ward-round tasks, reducing the workload of healthcare personnel. Autonomous following medical assistive robots provide companionship and assistance to patients through visual navigation and path planning technologies [12].

4) *Diagnostic Departments*

Assistive diagnostic robot utilizes high-resolution image, automatic image analysis, multi-modal image fusion with higher precision and faster rhythm of disease diagnosis. For instance, intestinal disease diagnosis system for colonoscopic medical robot integrates deep learning and computer vision technology to realize automatic intestinal polyp segmentation and classification [30]. Intelligent medical imagers like OCT (Optical Coherence Tomography) robots give the doctor of medical organization to have higher precision of imaging facilities so that they can identify disease diagnosis at high precision by knowing the tissue structure details.

5) *Special Environment Applications*

Service robots possess characteristics such as adaptability to special environments, convenience of remote medical operations, and completeness of medical data collection, enabling them to play a crucial role in special medical rescue scenarios. For example, medical transport robots in nuclear, biological, and chemical environments can safely transport the injured and sick in high-risk environments [13-15], reducing the exposure risk of rescue personnel. These robots employ special protective materials and designs, enabling normal operation in harsh conditions such as radiation and chemical contamination.

III. EXISTING PROBLEMS

A. *Technical Bottlenecks*

Precision and Stability: For some extremely complex surgeries like deep tumor resection in neurosurgery, stability and operational precision of the robotic arms still need to be improved. Complex hand motion or delicate operation still can not simulate the effects as manual rehabilitation. For example, for fine hand movement simulation, robotic arms' flexibility and precision still need to be improved. For some sophisticated work or accurate diagnosis, it is difficult to reach manual operation level precision yet. For instance, when applying in magnetically controlled interventional therapy, the accurate positioning and controlling of micro-robot still be influenced by the distribution of the magnetic field and in-vivo patients'

environment.

High Costs: Since the cost of conducting R&D and manufacturing high-end surgical robots is extremely high, the cost of such surgical robots is also expensive, and it's difficult to be widely available in primary hospitals. The cost of a unit Da Vinci surgical robot can be over ten million yuan.

Lack of Flexibility: Patient's need for rehabilitation, physical state and etc. vary greatly, and the existing rehabilitation robots cannot provide personal rehabilitation plans. Moreover, the control methods and ways of interaction for rehabilitation in special patient groups (such as severely paralyzed patients) still need improvement with current robots.

B. *Challenges in Clinical Application*

Doctor Training: Utilization of the robot surgery and rehab robots requires that doctors are endowed with high-tech, with rich experience, the training period is long and the cost is high. In China, currently no effective systematic operation training mechanism. The operation of the rehabilitation robot needs that the doctors should have a certain knowledge, experience and so on. Nevertheless, due to most of the doctors in hospitals lack of limited understanding of rehabilitation robots and using experience, the use of rehabilitation robot in hospital is not widespread.

Patient Acceptance: Some patients have doubts about medical robots, worrying about their safety and effectiveness [16]. Especially elderly patients and those with low education levels have lower acceptance of new technologies.

Ethical and Legal Issues: The widespread application of surgical robots has raised a series of ethical and legal issues, such as responsibility attribution [17-18], data security, and privacy protection. Currently, relevant laws and regulations are still imperfect.

IV. CONCLUSION

Medical robot, as a deep integration of artificial intelligence and robot technology in medical field, is gradually changing traditional medical model and improving the efficiency and quality of medical service. Even though the development technology of medical robot has still some distance to reach maturity in terms of technology maturity, controlling the cost and the legal issues and ethics, its application prospect is broad in surgery, rehabilitation, nursing, medical transportation in nuclear, biological and chemical environment, oral medicine and so on. For the future, with continuous technical development and stable industry development, the role of medical robot in the medical field will be even greater.

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